

Harrison Button

He/Him

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EXPERIENCE

• Rio Tinto Exploration

Technology Specialist (Acting) Perth, Australia. Jan. 2025 – Present

- Manage operation of LIBS core-scanning device and assist management of on-site laboratory.
- Use ioGAS and python to perform QAQC and analysis of LIBS-derived mineralogy and chemistry data, for use-cases in ore-body knowledge, exploration vectoring, and pre-feasibility studies for future Rio Tinto mines.
- Upskill staff (in Australia and Rwanda) on operation of core-scanning device, and analysis of and integration of LIBS results into logging, modelling, and machine learning workflows.

Geodata Scientist II Perth, Australia. Jan. 2023 – Dec. 2024

- Deployed machine learning model pipelines for production environments using Docker and AWS.
- Scaled-up internal R&D projects, which unlocked new capabilities in geophysical, geochemical, and mineralogical modelling for target generation and prioritization for the global exploration teams.
- Full-stack development of various data science apps, including automating CI/CD routines using GitHub Actions, integrating Databricks, and contributing to Python libraries for company-wide knowledge share.
- Helped manage a Posit Connect server with over 100 users from the regional exploration teams.

Geodata Scientist I Perth, Australia. Aug. 2021 – Dec. 2022

- Implemented machine learning methods for ore-body knowledge and target generation in global exploration contexts using Python and R.
- Automated GIS workflows using ArcPy and wrote custom Excel macros using VBA, saving hours of manual work per month.
- Produced and maintained multiple internal and external data science apps, built using Dash, Streamlit, and R Shiny.

Vacation Student, Geophysics Brisbane, Australia. Nov. 2020 – Aug. 2021

- Completed routine analysis of geophysical, geochemical, and petrophysical data for target generation in the Mt Isa and Pilbara regions.
- Compiled both project and state-wide data sets across Australia.
- Supervised airborne and ground geophysical surveys undertaken by geophysical contractors.

• The University of Queensland

Tutor, School of Mathematics and Physics Brisbane, Australia. Jul. 2017 – Nov. 2020

- Tutored a wide range of introductory and advanced courses to specialist and non-specialist students. Select courses include DATA7202 "Statistical Methods for Data Science", STAT3006 "Statistical Learning", and STAT3500 "Problems and Applications in Modern Statistics".

EDUCATION

• The University of Western Australia

Graduate Certificate in Data Science Jul. 2022 – Sep. 2023

• The University of Queensland

Bachelor of Science (Honours) in Exploration Geophysics Feb. 2019 – Nov. 2020

- Thesis title: Linear inversion of seismic refraction amplitudes for near-surface velocity control.

Bachelor of Mathematics / Science in Statistics and Geological Sciences (Extended) Feb. 2014 – Nov. 2018

PUBLICATIONS

- Button, H. & Whitbread, M. (2024). Building an integrated geochemistry interpretation app for the Paterson Province. Proceedings of the International Applied Geochemistry Symposium Meeting, 14-18 October 2024, Adelaide, Australia. <https://www.appliedgeochemists.org/aag-events/30th-iag-symposium-2024>

SKILLS

- **Programming:** Python, R, $\text{\LaTeX}$, SQL.
- **Software:** AWS, Docker, Git, GitHub Actions, Databricks, Jira/Confluence, Posit Connect, Linux.
- **Geoscientific software:** Oasis Montaj, ioGas, ModelVision, Geoscience Analyst, ArcGIS Pro (including ArcPy).